

Gitelman Syndrome as a Rare Etiology of Atrial Fibrillation: A Case of Successful Catheter Ablation Following Refractory Medical Treatment

Vasu Malhotra, DO¹, Omar Zuhdi, MD¹, Aqeel Khanani, DO¹, Usman Siddiqui, MD², Mayuri Patel, MD¹

¹Lakeland Regional Health, Lakeland, Florida, USA

Introduction

- Gitelman syndrome is a rare autosomal recessive renal tubular disorder (with a prevalence of 1:40,000) characterized by:
 - Hypokalemia
 - Hypomagnesemia
 - Hypocalciuria
 - Metabolic alkalosis
- These electrolyte imbalances can lead to a wide range of clinical manifestations, including fatigue, muscle weakness, and, in rare instances, arrhythmias and cardiac arrest.
- Atrial fibrillation (AF) in patients with GS is a serious manifestation resulting from these underlying chronic electrolyte imbalances, particularly hypokalemia and hypomagnesemia, which can cause delayed repolarization and increased propensity for arrhythmias.

Discussion

- A subset of individuals with obesity exhibit a Metabolically Healthy Obesity (MHO)– normal blood lipids, blood pressure, and glycemic control despite excessive adiposity.
- Multiple landmark studies, including the AMORIS study, INTERHEART study, Québec Cardiovascular Study, and Copenhagen General Population Study, have demonstrated that the ApoB/ApoA1 ratio can predict risk even in those with normal lipid profiles.
- Based on thresholds established in these studies, our patient's ApoB/ApoA1 ratio (0.72) places him in a moderate-risk category, reflecting an increased burden of atherogenic particles despite normal LDL-C.
- Favorable genetic traits such as IRS1 variants, familial hypobetalipoproteinemia, PCSK9 loss-of-function mutations, ANGPTL3 mutations, CETP deficiency, ABCA1 mutations, LPIN1 variants, and APOA5 variants may explain these findings.

Case Presentation

- The patient is a 53-year-old female with a history of poorly managed Gitelman syndrome, who presented with complaints of muscle cramping in both hands and feet.
- The patient was found to be in atrial fibrillation. Labs revealed severe hypokalemia and hypomagnesemia, which were the suspected etiology of her new-onset atrial fibrillation.
- Despite medical therapy aimed at correcting her electrolyte abnormalities, her arrhythmia persisted, and she remained symptomatic.
- The patient did have good response to initial management with pacemaker implantation.
- She subsequently was trialed on antiarrhythmic drugs such as sotalol, which she failed, and dronedarone, which she wanted to discontinue. The patient also refused oral anticoagulation for associated stroke prophylaxis.
- The patient underwent catheter ablation, which successfully resolved her symptoms.
- She has remained symptom-free in subsequent follow-up evaluations, one-year post-procedure.

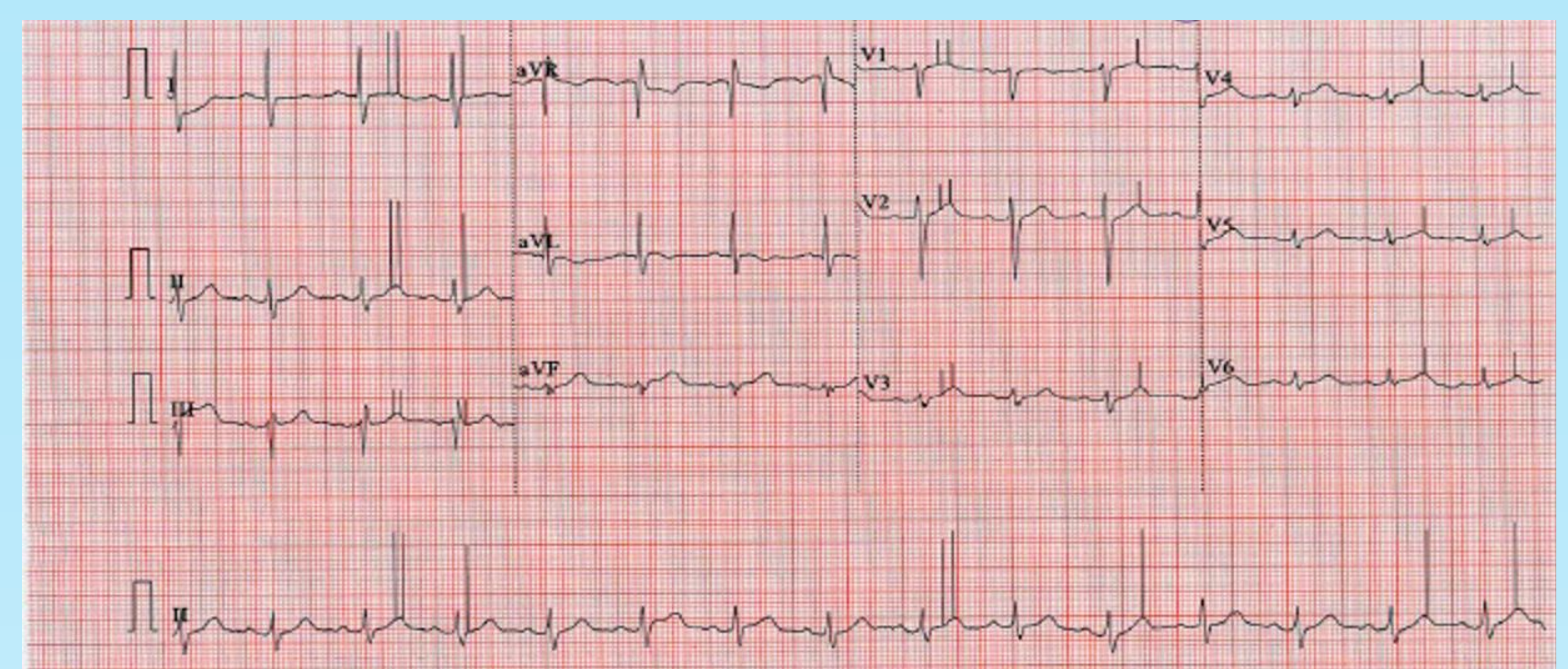


Figure 1 ECG Post-Catheter Ablation and Pacemaker Implantation Demonstrating Normalized Sinus Rhythm

The electrolyte abnormalities of Gitelman syndrome is a rarely reported etiology of atrial fibrillation requiring cardiac ablation in refractory cases.

